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HIGH-FREQUENCY TREATMENT HANDPIECE

Abstract

Proposed is a radio frequency treatment handpiece including a handpiece housing having a first containing space therein and a circuit board installed in the first containing space to operate the handpiece, and having a frontside thereof being open, an electrode tip having a nozzle portion that is detachably coupled to the handpiece housing and provided with an RF electrode for generating radio frequency waves, an electrode assembly installed in the first containing space, provided with a valve supply channel that receives a gaseous refrigerant from a second containing space and a main body, and includes a power distributor for applying electric power to the electrode tip and a solenoid valve configured to communicate with the valve supply channel and to control flow of the gaseous refrigerant, and an actuator coupled to the electrode assembly to vibrate the electrode assembly according to a control signal.

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Background/Summary

TECHNICAL FIELD

[0001] The present disclosure relates to a radio frequency treatment handpiece. More particularly, the present disclosure relates to a handpiece structure in which heat from various components, including electrodes, is removed by supplying a low-temperature gaseous refrigerant to prevent burns when radio frequency waves are generated on skin and allowing the gaseous refrigerant that has undergone heat exchange to flow inside the handpiece.

BACKGROUND ART

[0002] Human skin consists of layers of the stratum corneum, epidermis, dermis, and hypodermis, and the functions of the skin deteriorate due to aging and the action of ultraviolet rays.

[0003] The epidermis, which is the most superficial layer of the skin, is subdivided into several layers including the stratum corneum, stratum lucidum, stratum granulosum, stratum spinosum, and stratum basale depending on location and function, and plays important roles in protection, defense, and secretion.

[0004] The dermis is located beneath the epidermis and adjacent to the stratum basale, making up most of the skin. The dermis is composed of two layers: the papillary layer which contains moisture, proteins, carbohydrates, mucopolysaccharides, minerals, and inorganic salts in a jelly state and where capillaries related to blood circulation and lymphatic vessels carrying lymph are located; and the reticular layer which contains collagen, collagen fibers associated with skin wrinkles, elastin, elastic fibers that give skin elasticity, and a matrix (water reservoir).

[0005] In addition, the hypodermis is located between the dermis and muscles and bones, contains a large amount of fat, and constitutes the lowest layer of the skin. The hypodermis spreads evenly throughout the body, serves to maintain elasticity and acts as a buffer, absorbing external pressure and shock to prevent internal damage, as well as preventing heat loss and maintaining body temperature.

[0006] Meanwhile, as interest in skin increases, a variety of skin care devices related to skin treatments are being developed, and at the same time, many cosmetic dermatology hospitals and skin treatment clinics are emerging.

[0007] The most widely known skin treatment method recently is radio frequency (RF) therapy, in which radio frequency energy is applied to skin to induce local thermal damage within the skin to create microscopic wounds in order to promote skin regeneration from the epithelium to the dermis, and the microscopic wounds stimulate the release of cell growth factors to maximize natural healing and regeneration of the skin.

[0008] The above-described method delivers high heat energy to the skin by means of radio frequency, and thus there is a possibility that burns may occur in the epidermis. That is why a refrigerant is provided to a treatment area to cool the area to prevent burns. However, when water or a liquid refrigerant is used, refrigerant waste can be a problem because the refrigerant used for cooling is used only once and then discarded. Moreover, when water or a liquid refrigerant is used, a handpiece becomes heavy, putting a burden on the hands and wrists of an operator when performing a procedure for a long time.

[0009] On the other hand, in the case of radio frequency devices, when radio frequency waves are generated, considerable heat is produced not only in electrodes but also in components used to

drive the handpiece. Generally, a separate cooling device such as a fan is installed and used in a handpiece to remove such heat. In this case as well, the problem is that the handpiece becomes heavy as described above.

DISCLOSURE

Technical Problem

[0010] The present disclosure has been made to solve the above problems, and an objective of the present disclosure is to provide a cooling structure of a handpiece that simultaneously cools a treatment target area of a recipient and cools the inside of the handpiece.

Technical Solution

[0011] In order to achieve the above mentioned objective, there is provided a radio frequency treatment handpiece, including: a handpiece housing configured to have a first containing space therein and a circuit board installed in the first containing space so as to operate the handpiece, and have a frontside thereof being open; an electrode tip configured to have a nozzle portion that is detachably coupled to the handpiece housing and provided with an RF electrode for generating radio frequency waves by electric power applied thereto; an electrode assembly installed in the first containing space, provided with a valve supply channel that receives a gaseous refrigerant from a second containing space separate from the first containing space and a main body, and includes a power distributor for applying electric power to the electrode tip and a solenoid valve configured to communicate with the valve supply channel and to control flow of the gaseous refrigerant; and an actuator installed in a backside of the first containing space and coupled to the electrode assembly so as to vibrate the electrode assembly according to a control signal. The electrode assembly may include: a first body portion having a rear outlet that communicates the first containing space and the second containing space; [0012] a nozzle coupling portion coupled to an inner peripheral surface of the nozzle portion so as to close an open area; and a refrigerant supply channel configured to communicate with the valve supply channel so as to deliver a gaseous refrigerant to the electrode tip, and the solenoid valve may be provided between the refrigerant supply channel and the valve supply channel so as to control flow of the gaseous refrigerant according to a control signal.

[0013] In the radio frequency treatment handpiece according to an embodiment of the present disclosure, at least one through hole may be formed in a front outer wall of the handpiece housing, and on a side of the first body portion of the electrode assembly, a stepped portion that is inserted and joined into a fixing groove formed in the nozzle portion and a pressing portion that protrudes through the through hole of the handpiece housing may be provided.

[0014] In the radio frequency treatment handpiece according to an embodiment of the present disclosure, the refrigerant supply channel may be fixed on a central axis of the first body portion and communicates with the valve supply channel, the electrode assembly may further include: a second body portion having a channel through hole through which refrigerant supply channel passes, and formed to partition the first containing space and the second containing space; and a second flange extending outward from an end of the second body portion, and the power distributor may be inserted into and fixed to the second body portion.

[0015] In the radio frequency treatment handpiece according to an embodiment of the present disclosure, the power distributor may be installed to be fixed on the refrigerant supply channel, and a plurality of electrode leads may be formed on a surface of the power distributor to be electrically connected to the RF electrode.

[0016] In the radio frequency treatment handpiece according to an embodiment of the present disclosure, the second flange may be provided with a plurality of through holes on corresponding positions of a plurality of electrode leads, and the plurality of electrode leads may each penetrate the plurality of through holes and contact the RF electrode.

[0017] In the radio frequency treatment handpiece according to an embodiment of the present disclosure, the electrode tip may further include: a distributor that uniformly distributes the gaseous

refrigerant supplied from the refrigerant supply channel and supplies the gaseous refrigerant to the RF electrode; and a side outlet that communicates with the electrode assembly, so that the gaseous refrigerant that has exchanged heat with the RF electrode moves to the side outlet.

[0018] In the radio frequency treatment handpiece according to an embodiment of the present disclosure, a first outlet may be provided in an area of the first body portion to communicate with the side outlet and the second body portion, and a second outlet may be provided in the second body portion to communicate with the first outlet and the second containing space, so that the refrigerant heat exchange space, the second containing space, and the first containing space may be communicated in that order.

Advantageous Effects

[0019] According to a handpiece according to an embodiment of the present disclosure, since cooling of the internal space of a handpiece is achieved in addition to overall cooling of a treatment target area, it is possible to provide a lightweight handpiece without a separate cooling device, thereby reducing the fatigue and burden on an operator's hands and wrists during the procedure, enabling long-time operation and use.

Description

DESCRIPTION OF DRAWINGS

[0020] FIG. 1 is a cross-sectional view showing a cross-section of a handpiece according to an embodiment of the present disclosure;

[0021] FIG. 2 is an enlarged cross-sectional view showing a cross-section of the front side of a handpiece according to an embodiment of the present disclosure;

[0022] FIG. 3 is a cross-sectional view showing the handpiece structure with an electrode tip removed according to an embodiment of the present disclosure;

[0023] FIG. 4 is a top view showing the top surface of an electrode assembly according to an embodiment of the present disclosure;

[0024] FIG. 5 is a perspective view showing the electrode assembly viewed from the rear side according to an embodiment of the present disclosure; and

[0025] FIG. 6 is a conceptual diagram showing a supply flow path of a gaseous refrigerant and the movement of the gaseous refrigerant that has undergone heat exchange into the internal space of the handpiece by the handpiece structure according to an embodiment of the present disclosure.

BEST MODE

[0026] Similar reference numbers in the drawings refer to identical or similar functions across various aspects.

[0027] Hereinafter, specific details for carrying out the present disclosure will be described based on embodiments with reference to the drawings, and these embodiments are described in sufficient detail to enable those skilled in the art to practice the present disclosure.

[0028] It should be understood that the various embodiments of the present disclosure are different from one another but are not necessarily mutually exclusive. For example, specific shapes, structures, and characteristics described herein with respect to one embodiment may be implemented in other embodiments without departing from the spirit and scope of the present disclosure.

[0029] Therefore, the detailed description described below is not intended to be taken in a limited sense, and the scope of the present disclosure is limited solely by the appended claims, together with all equivalents to what those claims assert if properly described.

[0030] A handpiece **100** according to an embodiment of the present disclosure includes a handpiece housing **110**, an actuator **120**, an electrode tip **130**, and an electrode assembly **140**.

<Description of Components of Handpiece>

[0031] Hereinafter, the components of the handpiece **100** will be described with reference to FIGS. **1** to **5**.

[0032] The handpiece housing **110** has an open front and has a first containing space **110S** thereinside. The rear end of the first containing space **110S** is connected to a cable (not shown) connected to a main body (not shown).

[0033] In addition, a circuit board **112** for operating the handpiece is installed inside the handpiece housing **110**. The circuit board **112** is electrically connected to a plurality of wires, signal lines, etc. in the cable. The circuit board **112** is connected to various mechanical/electrical components such as the actuator **120**, an RF electrode **131**, or a solenoid valve **140V** in the handpiece **100**.

Accordingly, various control signals such as a control signal for radio frequency generation, opening and closing control of the solenoid valve **140V**, and vibration control for the electrode assembly **140** are received to control the overall operation of the handpiece.

[0034] The rear of the handpiece housing **110** is connected to the main body (not shown) through the cable.

[0035] In addition, at least one through hole **111** is formed on the outer wall of the front side of the handpiece housing **110**. The through hole **111** is for combination with the electrode assembly **140**, which will be described below.

[0036] Although the first containing space **110S** of the handpiece housing **110** is not divided into front and rear, for convenience of explanation, the first containing space **110S** will be described as being divided into a front containing space and a rear containing space.

[0037] The actuator **120** is installed in the rear area of the first containing space **110S**. In addition, a drive shaft of the actuator **120** is coupled to the rear of the electrode assembly **140** so that the electrode assembly may be moved in the forward and backward directions. Meanwhile, one end of a restoration spring **120S** is fixed to one surface of the handpiece housing **110**, and the other end of the restoration spring **120S** is fixed to the rear surface of the electrode assembly **140**.

[0038] Accordingly, when the actuator **120** operates to move the electrode assembly **140** in the forward and backward directions, in response to the displacement of the electrode assembly **140**, the restoration spring **120S** restores the electrode assembly **140** to the original position thereof. That is, the electrode assembly **140** may be vibrated in the forward and backward directions due to displacement by the actuator **120** and restoration by the restoration spring **120S**. At this time, the frequency may vary depending on the control signal used to operate the actuator **120**.

[0039] The electrode tip **130** has a tapered tube-shaped nozzle portion **130N** so as to be detachable from the handpiece housing **110**. The side wall of the nozzle portion **130N** is inserted to be at least partially positioned between the handpiece housing **110** and the electrode assembly **140**. In addition, at least one RF electrode **131** that receives power and generates radio frequencies is installed to close the open surface of the nozzle portion **130N**, and in this case, a predetermined distance is spaced between the back of the RF electrode **131** and a distributor **132** to form a predetermined refrigerant heat exchange space **130S**. Meanwhile, when the front of the RF electrode **131** comes in contact with the skin surface, the RF electrode **131** is capacitively connected to the skin surface. Depending on implementation, the RF electrode **131** may be made of a single electrode or may be provided as a combination of multiple electrodes.

[0040] The distributor **132** is mechanically connected to a refrigerant supply channel **140I** and fine channels are formed to provide a gaseous refrigerant to the entire rear area of the RF electrode **131**. In general, when supplying gaseous refrigerant to the RF electrode **131** through a single flow path, since only the rear area of the RF electrode **131** located on the same axis as the refrigerant supply channel **140I** is cooled intensively, a temperature gradient occurs in which the temperature difference becomes more severe as the distance from the central area of the RF electrode **131** increases. Thus, by providing the distributor **132**, the entire area on the back of the RF electrode **131** is cooled evenly.

[0041] In addition, although not shown in the drawings, a flexible power distributor for uniformly

providing power to the RF electrodes **131** or a plurality of electric pads corresponding to the number of RF electrodes **131** may be further provided on one side or inside the distributor **132**. The back of the flexible power distributor or the back of the electric pads is electrically connected to an electrode lead **145E** of a power distributor **145**.

[0042] A side outlet **133** is a fine channel provided on the outside of the distributor **132** and communicates with the electrode assembly **140**.

[0043] The electrode assembly **140** is a housing installed in front of the first containing space **110S** of the handpiece housing **110**, is formed with a second containing space **140S** having a separate space from the containing space **110S**, has a built-in power distributor for operating the handpiece, and a flow path is formed for the transfer of gaseous refrigerant.

[0044] The electrode assembly **140** is installed in front of the first containing space **110S** to deliver gaseous refrigerant delivered through a soft refrigerant supply tube (not shown) in communication with the main body to the electrode tip **130** along a predetermined flow path.

[0045] The electrode assembly **140** has a structure formed integrally with a first body portion **141** and a nozzle coupling portion **142**. In addition, a second body portion **143** and a second flange **144** are integrally formed and fixedly coupled to the first body portion **141**. For convenience of explanation, each part will be divided into components.

[0046] It is preferable that the first body portion **141** has a rear outlet **141H** formed in at least one area thereof that communicates with the first containing space **110S**, and that the front side of the first body portion **141** is closed or first body portion **141** has a closed structure that ensures airtightness. As a result, the first containing space **110S** and the second containing space **140S** are separated. It is desirable that the two spaces communicate only through the rear outlet **141H** so that a closed system in which gaseous refrigerant may circulate is formed inside the handpiece housing **110**. In addition, a first outlet **141O** is provided on one area of the first body portion **141** communicating with the side outlet **133** of the electrode tip **130** and the second body portion **143**.

[0047] The first body portion **141** may be deformed to some extent when pressed and is preferably made of a soft material such as electrically insulating plastic, but is not limited thereto.

[0048] Meanwhile, a stepped portion **141B** and a pressing portion **141P** are provided in the first body portion **141**. The stepped portion **141B** is formed on one side of the first body portion **141**. The step portion **141B** is inserted and fitted into a fixing groove formed in the nozzle portion **130N**. The pressing portion **141P** is also formed on one side of the first body portion **141**, and the pressing portion **141P** protrudes so as to penetrate a through hole **111** formed in the handpiece housing **110S**.

[0049] Accordingly, when a recipient **t** (a person receiving the treatment) presses the pressing portion **141P** after finishing use, the first body portion **141** is deformed and the stepped portion **141B** falls out from the fixing groove formed in the nozzle portion **130N**. As a result, the electrode tip **130** is separated from the handpiece housing **110**, and after separation, the recipient stops pressing the pressing portion **141P** to restore the first body portion **141** to the initial state thereof.

[0050] The refrigerant supply channel **140I** is provided on the second containing space **140S** and is installed on the central axis of the first body portion **141**. To be specific, the refrigerant supply channel **140I** is installed through a channel through hole **143H** formed in the longitudinal direction along the central axis of the second body portion **143**, which will be described later. One side of the refrigerant supply channel **140I** is in communication with the refrigerant heat exchange space **130S**, and the other side of the refrigerant supply channel **140I** is in mechanical communication with the solenoid valve **140V**.

[0051] The solenoid valve **140V** receives a valve opening/closing signal synchronized with a control signal for generating radio frequency from the main body (not shown) to generate radio frequency at the electrode tip **130** and open the solenoid valve **140V**. One side of the solenoid valve **140V** is mechanically communicated with the refrigerant supply channel **140I**, and the other side of the solenoid valve **140V** is mechanically communicated with the valve supply channel **140A**.

[0052] Meanwhile, one side of the valve supply channel **140A** is communicated with the solenoid

valve **140V**, and the other side of the valve supply channel **140A** is mechanically communicated with the refrigerant supply tube (not shown) to receive gaseous refrigerant from the main body (not shown).

[0053] Accordingly, the cold gaseous refrigerant provided from the main body (not shown) is delivered to the solenoid valve **140V** through the valve supply channel **140A**, and the gaseous refrigerant that has passed through the solenoid valve **140V** passes through the refrigerant supply channel **140I** and is provided to the refrigerant heat exchange space **130S** of the electrode tip **130**.

[0054] The nozzle coupling portion **142** is preferably configured to be coupled to the inner peripheral surface of the nozzle portion **130N** to close the open area. A space is created between the electrode tip **130** and the first body portion **141** by the nozzle coupling portion **142**, and this space is divided as a second containing space **140S'**. Meanwhile, a space of the first body portion **141** located behind the second body portion **143** will be described by dividing the space as the previously described second containing space **140S**. The reason for describing the two spaces separately is to provide convenience in explaining the flow of gaseous refrigerant later.

[0055] The second body portion **143** is installed on the first body portion **141** and is installed inside the second containing space (**140S**) and may be provided in a predetermined cylindrical shape. The channel through hole **143H** is formed in the longitudinal direction along the central axis of the second body portion **143** so that the refrigerant supply channel **140I** may be installed through the central axis of the second body portion **143**. In addition, a second outlet **143O** communicating with the first outlet **141O** and the second containing space **140S** is formed on the inner peripheral surface or the lower surface of the second body portion **143**.

[0056] The second flange **144** extends from one end of the second body portion **143** in an outward direction. A plurality of through holes **144H** are formed on the second flange **144** at positions corresponding to a plurality of electrode leads **145E**. In addition, a predetermined open groove **144O** is formed in the second flange **144** so that the stepped portion **141B** may be displaced in the direction of the central axis by the pressure of the pressing portion **141P**.

[0057] The power distributor **145** is fixedly installed on the refrigerant supply channel **140I** and is installed adjacent to the rear of the second flange **144**. The plurality of electrode leads **145E** are formed on one surface of the power distributor **145** and are electrically connected to the RF electrode **131** of the electrode tip **130**. Preferably, the electrode leads **145E** are provided to pass through the through holes **144H** formed on the second flange **144**. Accordingly, the electrode leads **145E** may be electrically connected by contacting the back of the flexible power distributor or the back of the electric pads located in the distributor **132** of the electrode tip **130** described above.

[0058] Hereinafter, the path along which the gaseous refrigerant flows will be briefly described with reference to FIG. 6.

<Description of Flow Path of Gaseous Refrigerant>

[0059] With reference to FIG. 6, the flow path of the gaseous refrigerant will be described in order.

[0060] 1. The gaseous refrigerant is supplied to the valve supply channel **140A** by a refrigerant supply means (not shown) installed in the main body outside the handpiece **100**. [main body->valve supply channel **140A**]

[0061] 2. The valve supply channel **140A** is mechanically communicated with the solenoid valve **140V**, so that when the solenoid valve **140V** is opened, the gaseous refrigerant is transferred from the valve supply channel **140A** to the refrigerant supply channel **140I**. [valve supply channel **140A**->solenoid valve **140V**->refrigerant supply channel **140I**]

[0062] 3. The refrigerant supply channel **140I** is mechanically communicated with the refrigerant heat exchange space **130S** of the electrode tip **130**, and transfers the gaseous refrigerant passing through the solenoid valve **140V** to the refrigerant heat exchange space **130S**. [refrigerant supply channel **140I**->refrigerant heat exchange space **130S**]

[0063] 4. Heat exchange occurs when the gaseous refrigerant comes into contact with the back of the RF electrode **131**. [0064] 5. The

gaseous refrigerant that has undergone heat exchange passes through the side outlet **133** and is delivered to the second containing space **140S'**. [refrigerant heat exchange space **130S**->side outlet

133->second containing space 140S'] [0065] 6. The gaseous refrigerant passing through the side outlet 133 is collected in the second containing space 140S, and the gaseous refrigerant is moved toward the second body portion 143 through the first outlet 141O. [second containing space 140S->first outlet 141O->second body portion 143] [0066] 7. The gaseous refrigerant is moved to the second containing space 140S through the second outlet 143O provided in the second body portion 143. [second body portion 143->second outlet 143O->second containing space 140S] [0067] 8. The gaseous refrigerant moves to the rear outlet 141H provided in the first body portion 141, thereby cooling the actuator 120. [second containing space 140S->rear outlet 141H->actuator 120] [0068] As described above, the present disclosure has been described with specific details such as specific components, limited embodiments, and drawings, but these are provided only to facilitate a more general understanding of the present disclosure, and the present disclosure is not limited to the above embodiments. Various modifications and variations may be made from the above description by those skilled in the art to which the present disclosure pertains. [0069] Therefore, the spirit of the present disclosure should not be limited to the described embodiments. Not only the scope of the patent claims described below, but also all things that are equivalent or have equivalent modifications to the scope of these claims fall within the scope of the present disclosure.

Claims

1. A radio frequency treatment handpiece, comprising: a handpiece housing configured to have a first containing space therein and a circuit board installed in the first containing space so as to operate the handpiece, and have a frontside thereof being open; an electrode tip configured to have a nozzle portion that is detachably coupled to the handpiece housing and provided with an RF electrode for generating radio frequency waves by electric power applied thereto; an electrode assembly installed in the first containing space, provided with a valve supply channel that receives a gaseous refrigerant from a second containing space separate from the first containing space and a main body, and includes a power distributor for applying electric power to the electrode tip and a solenoid valve configured to communicate with the valve supply channel and to control flow of the gaseous refrigerant; and an actuator installed in a backside of the first containing space and coupled to the electrode assembly so as to vibrate the electrode assembly according to a control signal, wherein the electrode assembly comprises: a first body portion having a rear outlet that communicates the first containing space and the second containing space; a nozzle coupling portion coupled to an inner peripheral surface of the nozzle portion so as to close an open area; and a refrigerant supply channel configured to communicate with the valve supply channel so as to deliver a gaseous refrigerant to the electrode tip, and the solenoid valve is provided between the refrigerant supply channel and the valve supply channel so as to control flow of the gaseous refrigerant according to a control signal.
2. The radio frequency treatment handpiece of claim 1, wherein at least one through hole is formed in a front outer wall of the handpiece housing, and on a side of the first body portion of the electrode assembly, a stepped portion that is inserted and joined into a fixing groove formed in the nozzle portion and a pressing portion that protrudes through the through hole of the handpiece housing are provided.
3. The radio frequency treatment handpiece of claim 1, wherein the refrigerant supply channel is fixed on a central axis of the first body portion and communicates with the valve supply channel, the electrode assembly further comprises: a second body portion having a channel through hole through which refrigerant supply channel passes, and formed to partition the first containing space and the second containing space; and a second flange extending outward from an end of the second body portion, and the power distributor is inserted into and fixed to the second body portion.
4. The radio frequency treatment handpiece of claim 3, wherein the power distributor is installed to

be fixed on the refrigerant supply channel, and a plurality of electrode leads are formed on a surface of the power distributor to be electrically connected to the RF electrode.

5. The radio frequency treatment handpiece of claim 3, wherein the second flange is provided with a plurality of through holes on corresponding positions of a plurality of electrode leads, and the plurality of electrode leads each penetrate the plurality of through holes and contact the RF electrode.

6. The radio frequency treatment handpiece of claim 3, wherein the electrode tip further comprises: a distributor that uniformly distributes the gaseous refrigerant supplied from the refrigerant supply channel and supplies the gaseous refrigerant to the RF electrode; and a side outlet that communicates with the electrode assembly, so that the gaseous refrigerant that has exchanged heat with the RF electrode moves to the side outlet.

7. The radio frequency treatment handpiece of claim 6, wherein a first outlet is provided in an area of the first body portion to communicate with the side outlet and the second body portion, and a second outlet is provided in the second body portion to communicate with the first outlet and the second containing space, so that the refrigerant heat exchange space, the second containing space, and the first containing space are communicated in that order.
